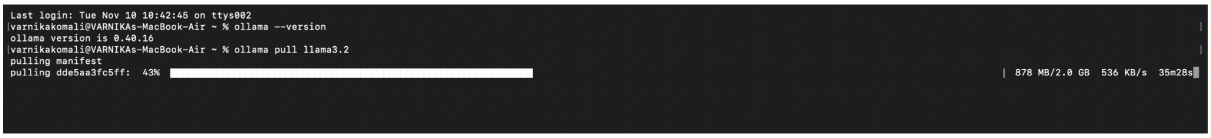
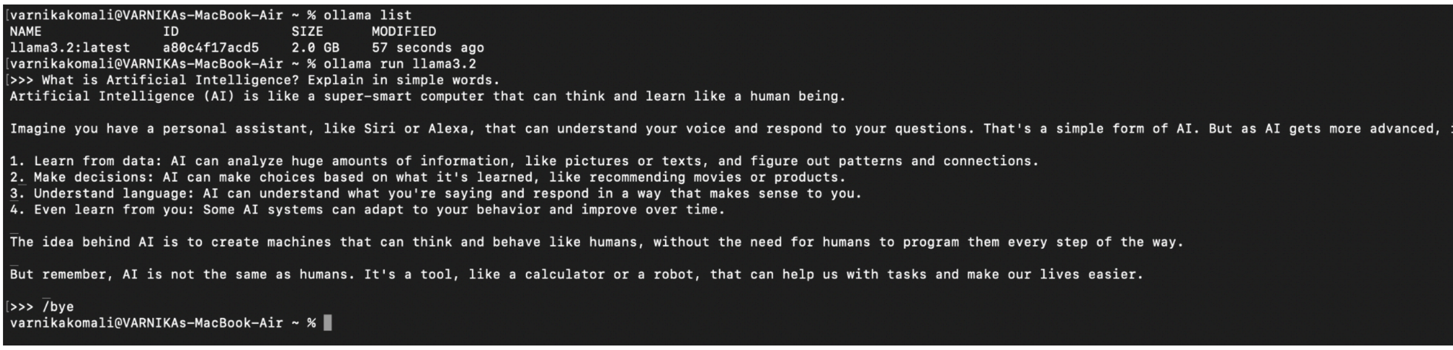
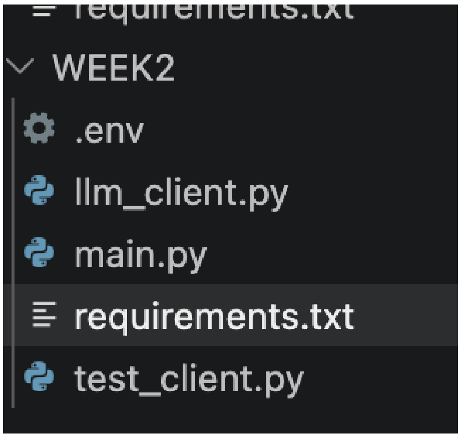
# 2520090010 KOMALI VARNIKA

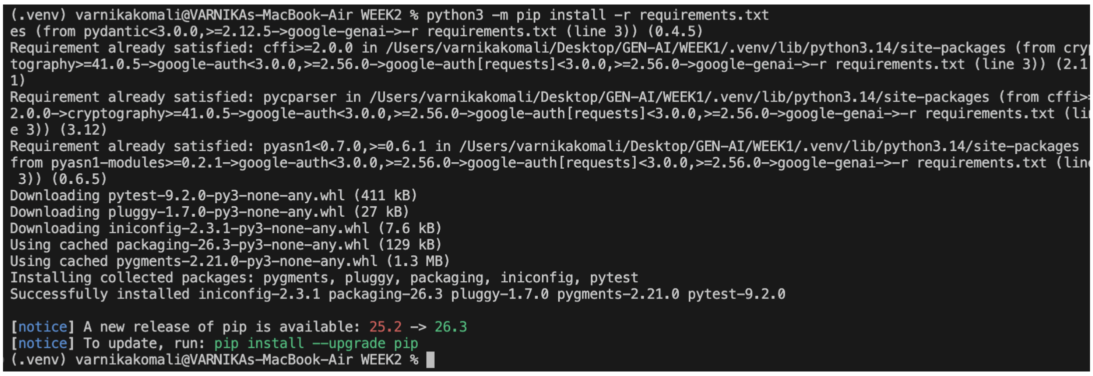


```
Last login: Tue Nov 10 10:42:45 on ttys002
[varnikakomali@VARNIKAs-MacBook-Air ~ % ollama --version
ollama version is 0.40.16
[varnikakomali@VARNIKAs-MacBook-Air ~ % ollama pull llama3.2
pulling manifest
pulling dde5aa3fc5ff:  43%                                                                   | 878 MB/2.0 GB  536 KB/s  35m28s
```

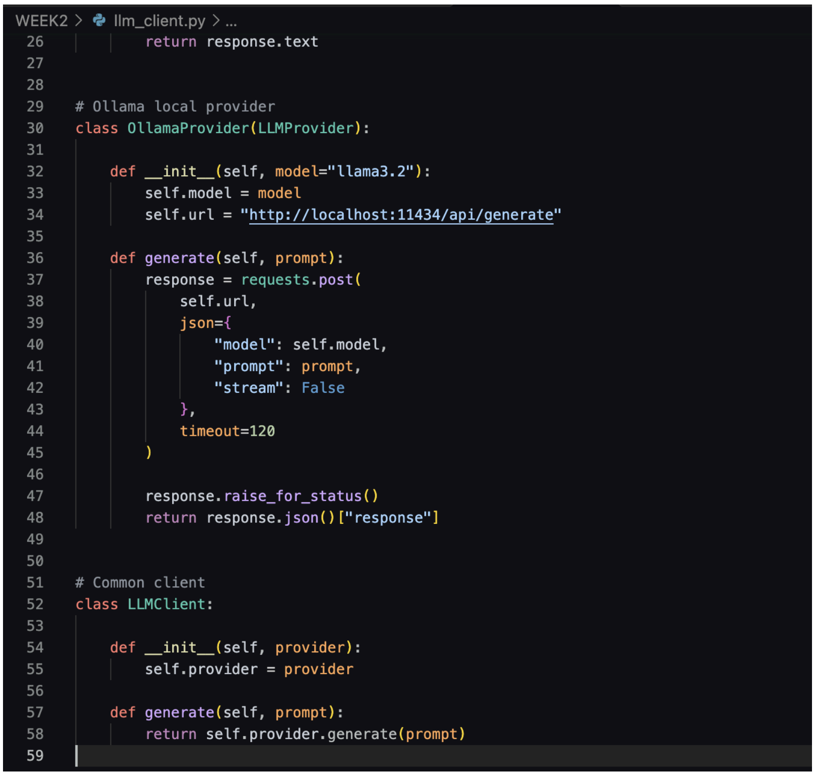


```
[varnikakomali@VARNIKAs-MacBook-Air ~ % ollama list
NAME               ID              SIZE      MODIFIED
llama3.2:latest    a80c4f17acd5    2.0 GB    57 seconds ago
[varnikakomali@VARNIKAs-MacBook-Air ~ % ollama run llama3.2
[>>> What is Artificial Intelligence? Explain in simple words.
Artificial Intelligence (AI) is like a super-smart computer that can think and learn like a human being.

Imagine you have a personal assistant, like Siri or Alexa, that can understand your voice and respond to your questions. That's a simple form of AI. But as AI gets more advanced,

1. Learn from data: AI can analyze huge amounts of information, like pictures or texts, and figure out patterns and connections.
2. Make decisions: AI can make choices based on what it's learned, like recommending movies or products.
3. Understand language: AI can understand what you're saying and respond in a way that makes sense to you.
4. Even learn from you: Some AI systems can adapt to your behavior and improve over time.

The idea behind AI is to create machines that can think and behave like humans, without the need for humans to program them every step of the way.

But remember, AI is not the same as humans. It's a tool, like a calculator or a robot, that can help us with tasks and make our lives easier.

[>>> /bye
varnikakomali@VARNIKAs-MacBook-Air ~ %
```

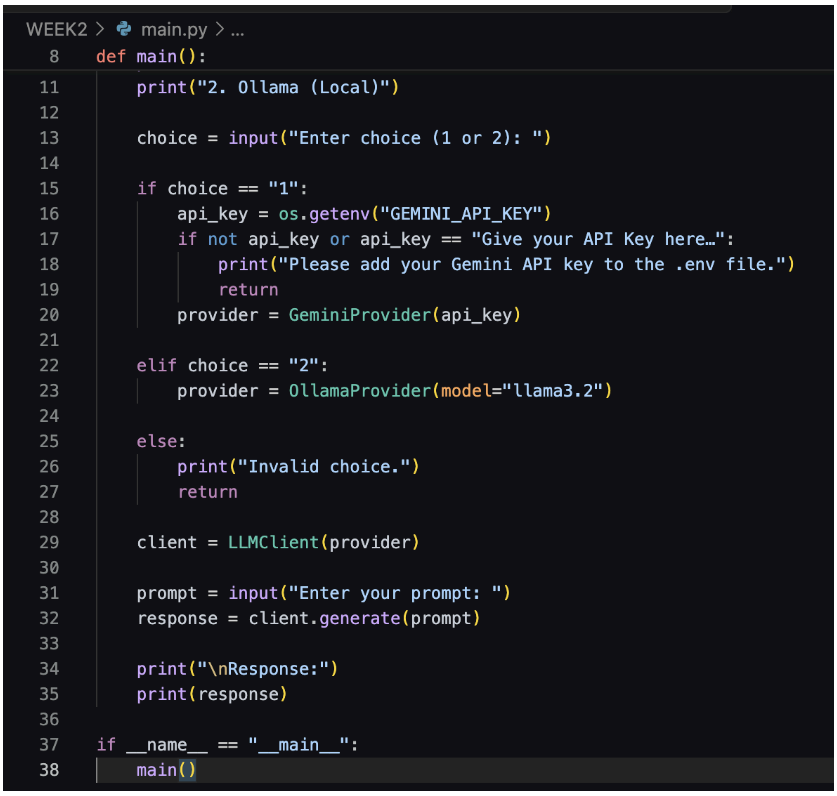




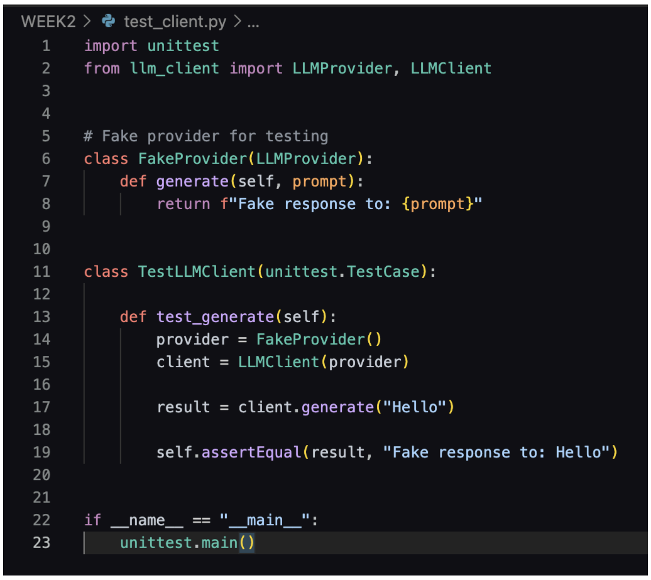
```
(.venv) varnikakomali@VARNIKAs-MacBook-Air WEEK2 % python3 -m pip install -r requirements.txt
es (from pydantic<3.0.0,>=2.12.5->google-genai->-r requirements.txt (line 3)) (0.4.5)
Requirement already satisfied: cffi>=2.0.0 in /Users/varnikakomali/Desktop/GEN-AI/WEEK1/.venv/lib/python3.14/site-packages (from cry
tography>=41.0.5->google-auth<3.0.0,>=2.56.0->google-auth[requests]<3.0.0,>=2.56.0->google-genai->-r requirements.txt (line 3)) (2.1
1)
Requirement already satisfied: pycparser in /Users/varnikakomali/Desktop/GEN-AI/WEEK1/.venv/lib/python3.14/site-packages (from cffi>
2.0.0->cryptography>=41.0.5->google-auth<3.0.0,>=2.56.0->google-auth[requests]<3.0.0,>=2.56.0->google-genai->-r requirements.txt (li
e 3)) (3.12)
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /Users/varnikakomali/Desktop/GEN-AI/WEEK1/.venv/lib/python3.14/site-packages
from pyasn1-modules>=0.2.1->google-auth<3.0.0,>=2.56.0->google-auth[requests]<3.0.0,>=2.56.0->google-genai->-r requirements.txt (lin
 3)) (0.6.5)
Downloading pytest-9.2.0-py3-none-any.whl (411 kB)
Downloading pluggy-1.7.0-py3-none-any.whl (27 kB)
Downloading iniconfig-2.3.1-py3-none-any.whl (7.6 kB)
Using cached packaging-26.3-py3-none-any.whl (129 kB)
Using cached pygments-2.21.0-py3-none-any.whl (1.3 MB)
Installing collected packages: pygments, pluggy, packaging, iniconfig, pytest
Successfully installed iniconfig-2.3.1 packaging-26.3 pluggy-1.7.0 pygments-2.21.0 pytest-9.2.0

[notice] A new release of pip is available: 25.2 -> 26.3
[notice] To update, run: pip install --upgrade pip
(.venv) varnikakomali@VARNIKAs-MacBook-Air WEEK2 %
```



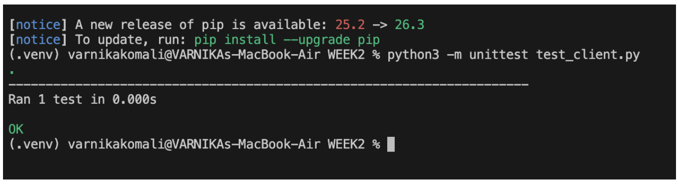
```python
        return response.text


# Ollama local provider
class OllamaProvider(LLMProvider):

    def __init__(self, model="llama3.2"):
        self.model = model
        self.url = "http://localhost:11434/api/generate"

    def generate(self, prompt):
        response = requests.post(
            self.url,
            json={
                "model": self.model,
                "prompt": prompt,
                "stream": False
            },
            timeout=120
        )

        response.raise_for_status()
        return response.json()["response"]


# Common client
class LLMClient:

    def __init__(self, provider):
        self.provider = provider

    def generate(self, prompt):
        return self.provider.generate(prompt)

```



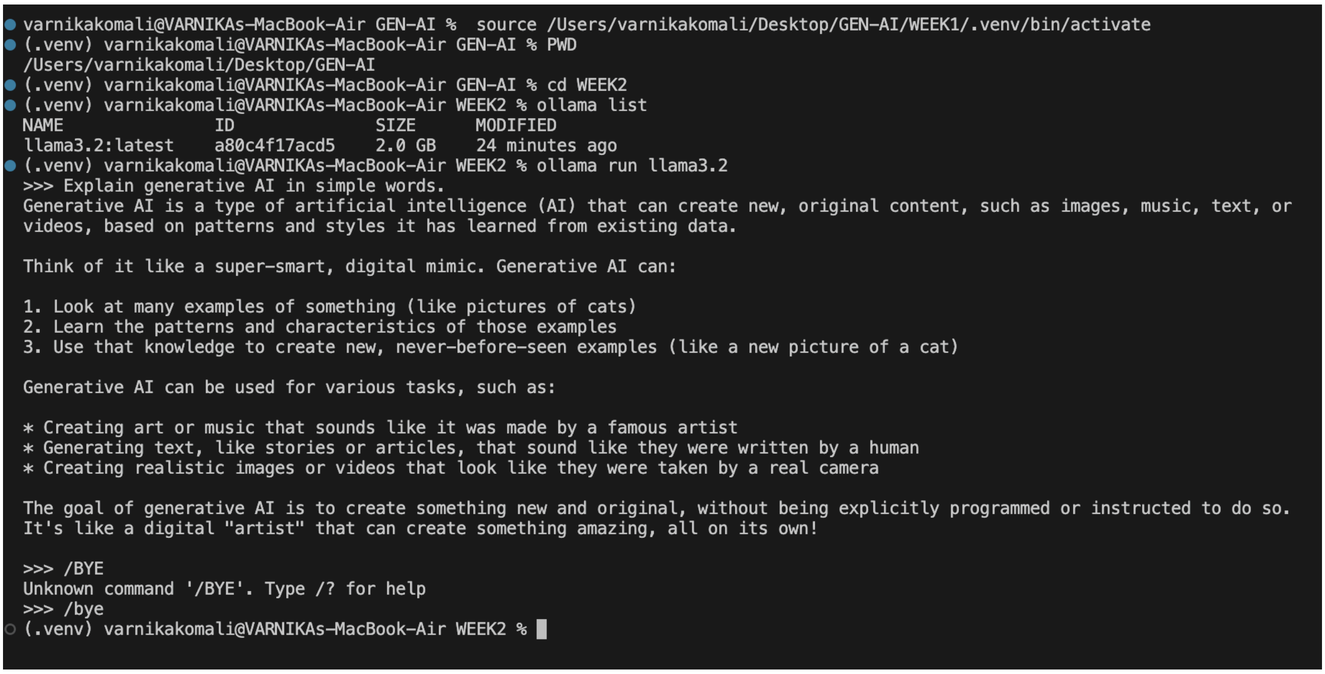
```python
def main():
    print("2. Ollama (Local)")

    choice = input("Enter choice (1 or 2): ")

    if choice == "1":
        api_key = os.getenv("GEMINI_API_KEY")
        if not api_key or api_key == "Give your API Key here…":
            print("Please add your Gemini API key to the .env file.")
            return
        provider = GeminiProvider(api_key)

    elif choice == "2":
        provider = OllamaProvider(model="llama3.2")

    else:
        print("Invalid choice.")
        return

    client = LLMClient(provider)

    prompt = input("Enter your prompt: ")
    response = client.generate(prompt)

    print("\nResponse:")
    print(response)

if __name__ == "__main__":
    main()
```



```python
import unittest
from llm_client import LLMProvider, LLMClient


# Fake provider for testing
class FakeProvider(LLMProvider):
    def generate(self, prompt):
        return f"Fake response to: {prompt}"


class TestLLMClient(unittest.TestCase):

    def test_generate(self):
        provider = FakeProvider()
        client = LLMClient(provider)

        result = client.generate("Hello")

        self.assertEqual(result, "Fake response to: Hello")


if __name__ == "__main__":
    unittest.main()
```



```
[notice] A new release of pip is available: 25.2 -> 26.3
[notice] To update, run: pip install --upgrade pip
(.venv) varnikakomali@VARNIKAs-MacBook-Air WEEK2 % python3 -m unittest test_client.py
.
----------------------------------------------------------------------
Ran 1 test in 0.000s

OK
(.venv) varnikakomali@VARNIKAs-MacBook-Air WEEK2 % 
```



```
varnikakomali@VARNIKAs-MacBook-Air GEN-AI %  source /Users/varnikakomali/Desktop/GEN-AI/WEEK1/.venv/bin/activate
(.venv) varnikakomali@VARNIKAs-MacBook-Air GEN-AI % PWD
/Users/varnikakomali/Desktop/GEN-AI
(.venv) varnikakomali@VARNIKAs-MacBook-Air GEN-AI % cd WEEK2
(.venv) varnikakomali@VARNIKAs-MacBook-Air WEEK2 % ollama list
NAME               ID              SIZE      MODIFIED
llama3.2:latest    a80c4f17acd5    2.0 GB    24 minutes ago
(.venv) varnikakomali@VARNIKAs-MacBook-Air WEEK2 % ollama run llama3.2
>>> Explain generative AI in simple words.
Generative AI is a type of artificial intelligence (AI) that can create new, original content, such as images, music, text, or
videos, based on patterns and styles it has learned from existing data.

Think of it like a super-smart, digital mimic. Generative AI can:

1. Look at many examples of something (like pictures of cats)
2. Learn the patterns and characteristics of those examples
3. Use that knowledge to create new, never-before-seen examples (like a new picture of a cat)

Generative AI can be used for various tasks, such as:

* Creating art or music that sounds like it was made by a famous artist
* Generating text, like stories or articles, that sound like they were written by a human
* Creating realistic images or videos that look like they were taken by a real camera

The goal of generative AI is to create something new and original, without being explicitly programmed or instructed to do so.
It's like a digital "artist" that can create something amazing, all on its own!

>>> /BYE
Unknown command '/BYE'. Type /? for help
>>> /bye
(.venv) varnikakomali@VARNIKAs-MacBook-Air WEEK2 % 
```